Stat 240: Lab 00

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AUTHOR
Peter Khrissanov

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This introductory lab is to ensure you have the computing environment required for the rest of the semester set up. If you have taken a previous class involving computing in R much of this will be familiar. If not, this should still be straightforward.

This lab will be graded out of 1 point, to ensure you are familiar with the submission process for future problem sets.

Problem 1

- To complete the lab you will need to ensure you have installed R and RStudio. See the installation instructions in the files section of Canvas.

Problem 2

- Next you should go to the `lab_00` folder in canvas and download this file `lab_00.qmd`. **Save this file somewhere specific on your computer**. For example, create a folder for stat 240 and save it there.

We will talk a lot more about saving and organiizing files in future weeks, but this is just a very quick way to get started. (If you know what an RStudio project is you can set one up for the class before you do this. If not we will discuss it soon.)

Now open this file you have saved in RStudio. You will see little grayed out chunks corresponding to code, for example.

```r
num_1 <- 101
num_2 <- 303

num_1 + num_2
```

```
[1] 404
```

If you press the green play button this code will run and show you the result. You can create these code chunks with the shortcut `Cmd/Ctrl + Option/Alt + i`.

To Do. Create a code chunk and do a small simple calculation here.

```
word1 <- 3
word2 <- 2

word1+word2
```

```
[1] 5
```

Problem 3

Now we will see how to read in a simple data file into R, as seen in the first class. From canvas download `lab_00_data.csv` and **save it in the same folder as the qmd file you previously downloaded**.

You should then be able to run the following code, which will read this data into R and show you the first 6 rows of data.

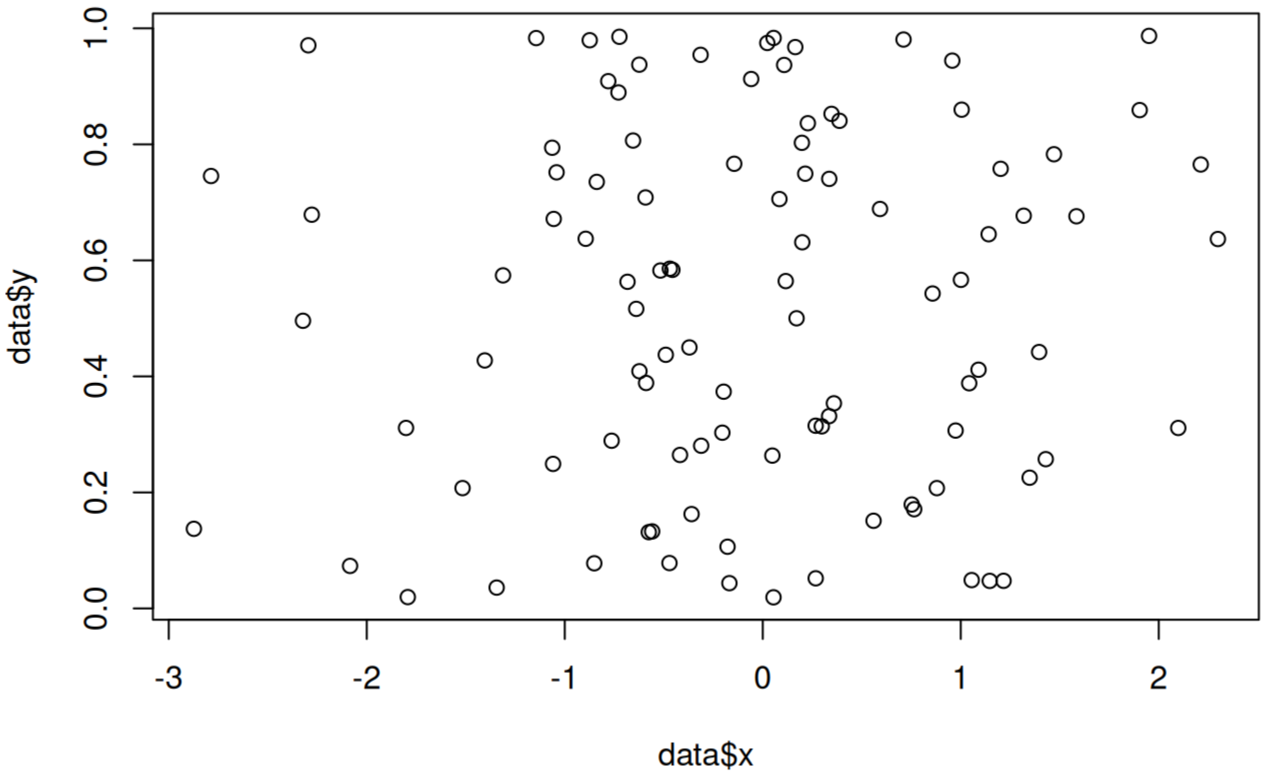
```
data <- read.csv("lab_00_data.csv")
head(data)
```

```
           x          y
1  1.1463225 0.04744809
2 -0.7632853 0.28910432
3 -0.5920996 0.70847406
4 -2.3221966 0.49588925
5 -0.5167792 0.58252126
6  0.3592260 0.35358959
```

Problem 4

Finally, we will create a very simple plot using this data, by running the following code. You will need to have read the data in in the previous step for this code to run!

```
plot(data$x, data$y)
```



Problem 5

To Do. Now create a pdf of your version of this document.

To do this you should:

- Modify the author section above, adding your name
- Render the quarto document, using the Render button at the top of the document.

Then, this will `Render` the document, creating a file with the same name but now with the `html` extension. It will also open in the `Viewer` tab beside files, where you can see what it looks like.

You can also open this `html` file by clicking on it, and opening it with your web browser.

When you render a file, code written inside a code chunk will be run and the output will be placed in the file created. This also means that if you write code which does not run properly, it will cause an error and RStudio will not be able to knit your file.

This is almost in the format we want to submit, except we need to convert it to `pdf`. To do this, when it is open in the browser you should `Print it to PDF`, saving it with the correct name.

This pdf is the file you should then submit to crowdmark for both labs and computer based exams in this course. This lab is to ensure you have this set up correctly for the rest of the semester!